

```
#include <systemc.h>

SC_MODULE(priority_interrupt) {

    sc_event low_int, high_int;

    void low_ISR() {
        while(true) {
            wait(low_int);
            cout << "Low Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(3, SC_SEC);
        }
    }

    void high_ISR() {
        while(true) {
            wait(high_int);
            cout << "High Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    void generator() {
        wait(1, SC_SEC);
    }
};
```

[Wrote 52 lines]

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```
praveenkumar@gmail:~$ nano Exp29.cpp
praveenkumar@gmail:~$ g++ -o Exp29 Exp29.cpp -lsystemc
praveenkumar@gmail:~$ ./Exp29
```

```
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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```

```
Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
praveenkumar@gmail:~$
```